



Modeling nonlinear scattering of phospholipid-coated microbubbles in elastic media

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ABSTRACT

Microbubble contrast agents are well-known in the biomedical field for their ability to enhance the contrast in ultrasound imaging and to facilitate targeted drug delivery within the body. Upon injection into the vasculature, these bubbles are strongly influenced by the viscoelastic properties of surrounding soft tissues. Here, we simulate the scattered pressure from a single phospholipid-coated microbubble placed in a uniform viscoelastic medium. We use a modified Rayleigh–Plesset equation that incorporates the viscoelastic response of the phospholipid shell and a Kelvin–Voigt model to represent tissue viscoelasticity. Our simulations aim to investigate the relationship between the viscoelasticity of the medium and that of the bubble shell, and its influence on bubble dynamics as a function of driving pressure, bubble radius, and medium elastic modulus. Results show that the frequency of maximum scatter response increases with the elastic modulus of the medium. While the radial excursion of a 2.0 μm radius bubble can decrease by as much as 45% in a medium with an elastic modulus of 200 kPa as compared to that in water, the scattered pressure can increase by 50%. In contrast, the change in harmonic scattering is negligible: simulations predict a increase of only 5% for a 2.0 μm bubble when increasing the medium elasticity from 0 to 200 kPa at a driving pressure of 120 kPa. Furthermore, subharmonics generated by driving the bubble at its resonance frequency (TR) are comparable to the subharmonics generated at twice the resonance frequency at lower driving pressures. For elevated pressures the TR/T2R subharmonic ratio increases with increasing elastic modulus. This is in sharp contrast with the resonance behavior of a microbubble in water. These results bear consequences for strategies that exploit the fundamental, harmonic, and subharmonic response of microbubble contrast agents.

1. Introduction

The healthcare sector is eagerly searching for imaging modalities that are both safe and cost-effective for diagnostic and therapeutic purposes. Ultrasound (US) has proven to be a reliable imaging method for quantifying (microvascular) blood flow and is being extensively investigated for therapeutic applications in lithotripsy [1], histotripsy [2], blood–brain barrier opening [3,4], or immunotherapy [5]. This modality often relies on the use of contrast agents to visualize the blood pool or to induce the therapeutic effects. In contrast-enhanced ultrasound imaging, tracking contrast intensity or speckle movement offers spatio-temporally resolved information about organ perfusion [6,7], the state of stenotic lesions and aneurysm rupture risks [8–10], or tumor malignancy [11]. In therapy, bubbles are the main mediators for therapeutic action on tissues by exerting localized mechanical stress on cells [12–15].

Ultrasound contrast agents consist of microbubbles that typically have a radius of 1 to 5 μm . To prevent the rapid dissolution of the gas core, the bubble is encapsulated in a biocompatible protective shell, typically consisting of phospholipids. Furthermore, the bubbles are often filled with a heavy, low-solubility, and inert gas, such as C_4F_{10} , to prolong their circulation time [16]. When exposed to ultrasound in the MHz frequency range, these microbubbles undergo volumetric oscillations and exhibit an acoustic scattering cross-section that is up to nine orders of magnitude larger than that of the similarly sized red blood cells. Bubbles can be driven into resonance near their natural frequency, which further increases their scattering cross-section by one order of magnitude [17,18].

The scattering strength of the microbubbles is key to their widespread use as ultrasound contrast agents. Furthermore, bubble dynamics are nonlinear, and the resulting harmonic components of the acoustic

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echo can be utilized to further enhance the contrast-to-tissue ratio in ultrasound imaging by suppressing the fundamental tissue echoes. These nonlinearities arise from both intrinsic bubble dynamics and extrinsic contributions, e.g., the (nonlinear) mechanics of the coating, and translate into the generation of second harmonics [19], ultraharmonics [20], and subharmonics [21–23].

Since understanding and predicting these nonlinearities is a prerequisite for the effective use of bubbles, many studies have sought to model bubble dynamics, initially for the first-generation ultrasound contrast agents such as Albunex and Optison [24–26]. Following this pioneering work, the introduction of so-called second-generation microbubbles coated with phospholipids led to new modeling efforts [27–29], with a particular focus on the viscoelastic behavior of the lipid shell. Notably, Marmottant et al. [29] developed an ad-hoc model to describe the surface tension of the lipid-coated microbubble as a function of surface area, where the bubble surface can periodically buckle upon compression and rupture upon expansion. In the intermediate state, the microbubble coating behaves elastically and is characterized by a finite surface elasticity. This highly nonlinear response during bubble vibrations strongly affects the bubble dynamics and results in a pressure-dependent resonance frequency that is high for low driving pressure amplitudes and nearly decreases to that of a free bubble for high driving pressure [30,31]. Resonant oscillations also lead to increased nonlinear scattering [32].

The dynamics of microbubbles is also influenced by the mechanical properties of the surrounding medium. The presence of a soft, viscoelastic medium significantly affects both the linear and nonlinear behavior of the bubbles [33,34]. Prosperetti [33] studied the linear and nonlinear oscillations of gas bubbles in various media by introducing a new pressure term to a modified Rayleigh–Plesset equation so as to account for the non-Newtonian behavior of liquids and for mass exchange at the bubble interface. Furthermore, a series of analytical and numerical studies have explored cavitation in soft materials. For example, Allen and Roy [35,36] have incorporated viscoelastic constitutive equations into the bubble dynamics equations, highlighting the impact of both linear and nonlinear medium deformations on microbubble dynamics. Gaudron et al. [37] have investigated the effects of nonlinear elasticity on microbubble collapse using large deformation elastic models. More recently, Oratis et al. [38] have proposed a generalized model that accounts for arbitrary linear viscoelastic models using the relaxation function. **Since these models quickly become quite complex to handle, a simplified model, such as the linear Kelvin–Voigt model, is often used as constitutive model [39]. It predominantly applies to solid-like materials in a narrow excitation bandwidth and, by extension, to soft tissues that generally recover their original configuration after the stress is removed.**

While the aforementioned models have shown potential in predicting experimental outcomes and improving measurement accuracy, most studies on bubbles in viscoelastic materials have focused on examining the effects of therapeutic ultrasound where bubbles are driven at high pressures, e.g., for high-intensity focused ultrasound (HIFU) applications and cavitation [1,40,41]. Only a few studies have investigated the diagnostic imaging regime which uses much lower driving pressures [42–44]. Our understanding of the nonlinear behavior of bubbles in viscoelastic media is thus still incomplete, and in particular the role of the combined effects of shell and medium viscoelasticity on the bubble dynamics and the consequences on the scattered ultrasound signals are still largely unexplored.

Based on the well-established understanding of phospholipid shell behavior in Newtonian liquids [45], **this study examines the interplay between the shell and a Kelvin–Voigt viscoelastic medium using numerical simulations.** We aim to explore the linear and nonlinear scattered pressure of a **phospholipid-coated microbubble** as a function of the driving pressure and frequency of the ultrasound wave and as a function of the elastic modulus of the surrounding medium.

2. Theory

2.1. Bubble dynamics

In the linear regime of deformation of a **Kelvin–Voigt viscoelastic medium, the modified Rayleigh–Plesset (RP) equation** of motion for a spherically symmetric phospholipid-coated gas bubble can be expressed as follows [34]:

$$\rho_m \left(R\ddot{R} + \frac{3}{2}\dot{R}^2 \right) = \left(P_0 + \frac{2\sigma(R_0)}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\gamma} - P_0 - P_{ac} - \frac{2\sigma(R)}{R} - \frac{4\kappa_s \dot{R}}{R^2} - \frac{4\mu \dot{R}}{R} - \frac{4G}{3} \left(\frac{R^3 - R_0^3}{R^3} \right). \quad (1)$$

Here, ρ_m is the density of the medium, and R is the radius of the bubble. The single and double overdots represent the first and second time derivatives, respectively. The first right-hand-side term describes the gas pressure inside the bubble, where γ is the polytropic exponent of the gas and R_0 is the bubble radius at ambient pressure, P_0 , and P_{ac} is the acoustic driving pressure. The surface tension of the microbubble $\sigma(R)$ is size-dependent [29,45]. Finally, κ_s is the dilatational shell viscosity and the last two terms of Eq. (1) describe the surrounding medium which is represented by a Kelvin–Voigt model [34]. Its viscosity and elastic modulus are denoted by μ and G , respectively.

Eq. (1) can be linearized by assuming small oscillation amplitudes around the equilibrium radius, i.e., using $R = R_0(1 + x(t))$, where $x(t) \ll 1$. This yields:

$$\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2x = -\frac{P_{ac}}{\rho_m R_0^2} \sin(\omega t), \quad (2)$$

the classic equation for a driven harmonic oscillator where

$$\omega_0 = 2\pi f_0 = \frac{1}{R_0} \sqrt{\frac{1}{\rho_m} \left(3\gamma P_0 + (3\gamma - 1) \frac{2\sigma(R_0)}{R_0} + \frac{4\chi}{R_0} + 4G \right)} \quad (3)$$

is the natural angular frequency of the bubble. $\sigma(R_0)$ is the surface tension of the bubble at rest and χ is the shell elasticity [29,46]. In Eq. (2), ζ is the damping ratio that contains dissipative terms including acoustic re-radiation, medium viscosity, thermodynamic losses, and shell viscous dissipation. In the following we will simply refer to ζ as the total linear damping ratio that encompasses all these terms whose expression we do not further specify.

Using the complex notation represented by the underline, where $x = \text{real}(\underline{x}e^{j\omega t})$, the linearized oscillation amplitude is given by:

$$\left| \frac{\underline{x}}{\underline{P_{ac}}} \right| = \frac{\frac{1}{\rho_m R_0^2}}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\zeta\omega\omega_0)^2}}, \quad (4)$$

The resonance frequency ω_r can then be inferred from finding the frequency of maximum oscillation amplitude from the derivative of Eq. (4), which then writes:

$$\omega_r = \omega_0 \sqrt{1 - 2\zeta^2}. \quad (5)$$

Inserting Eq. (5) into Eq. (4) leads to the maximum oscillation amplitude at resonance:

$$\max \left(\left| \frac{\underline{x}}{\underline{P_{ac}}} \right| \right) = \frac{1}{2\rho_m R_0^2 \omega_0^2 \zeta \sqrt{1 - \zeta^2}}, \quad (6)$$

which decays quadratically with the natural frequency of the bubbles. In addition, shell and viscous damping ratios vary with ω_0^{-1} [46]. Using Eq. (3), considering a small damping ratio ($\zeta \ll 1$), the maximum oscillation amplitude scales with the inverse of ω_0 and, subsequently:

$$\max \left(\left| \frac{\underline{x}}{\underline{P_{ac}}} \right| \right) \propto \frac{1}{R_0} \frac{\sqrt{\rho_m}}{\sqrt{3\gamma P_0 + (3\gamma - 1) \frac{2\sigma(R_0)}{R_0} + \frac{4\chi}{R_0} + 4G}}, \quad (7)$$

which then decreases with the square root of the elastic modulus.

2.2. Scattered pressure

The scattered pressure from a microbubble can be obtained from the **momentum conservation equation in a weakly compressible medium** [47,48]:

$$P_s \left(r, t + \frac{r}{c} \right) = \frac{\rho_m R}{r} (2\dot{R}^2 + R\ddot{R}), \quad (8)$$

where c is the speed of sound. Note that R , $\dot{R}$, and $\ddot{R}$ are functions of time. Linearization in the context of harmonic oscillations then yields:

$$P_s \left(r, t + \frac{r}{c} \right) \approx \frac{\rho_m R_0^3}{r} \ddot{x} \approx -\omega^2 \frac{\rho_m R_0^3}{r} x. \quad (9)$$

Using the behavior of x described by Eq. (4):

$$\left| \frac{P_s}{P_{ac}} \right| \approx \frac{R_0}{r} \frac{\omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\zeta\omega\omega_0)^2}}, \quad (10)$$

Calculating the maximum of Eq. (10) provides the frequency of maximum response in terms of scattering:

$$\omega_{m,s} = \frac{\omega_0}{\sqrt{1 - 2\zeta^2}}, \quad (11)$$

which differs from the frequency of maximum response for the oscillation amplitude ω_r , see Eq. (4). Inserting Eq. (11) into Eq. (10) leads to the maximum scatter response:

$$\max \left(\left| \frac{P_s(r)}{P_{ac}} \right| \right) = \frac{R_0}{2r\zeta\sqrt{1 - \zeta^2}}. \quad (12)$$

Using the same arguments as for the amplitude of oscillations, and in the limit of small damping:

$$\max \left(\left| \frac{P_s(r)}{P_{ac}} \right| \right) \propto \sqrt{3\gamma P_0 + (3\gamma - 1) \frac{2\sigma(R_0)}{R_0} + \frac{4\chi}{R_0} + 4G}. \quad (13)$$

Thus, in contrast to the maximum amplitude of oscillation which decreases with increasing medium stiffness, see Eq. (7), the maximum scattered pressure increases with increasing stiffness of the medium, see Eq. (13).

3. Materials and methods

A schematic of the modeled system is shown in Fig. 1a: a phospholipid-coated microbubble filled with C_4F_{10} is placed in an infinite viscoelastic medium with elastic modulus G and viscosity μ . The bubble is insonified by a pressure wave (parameters listed in Table 1), and the scattered pressure is evaluated at a distance of $r = 1$ cm.

3.1. Numerical procedures

We solve Eq. (1) in MATLAB® using the ode45 ordinary differential equation (ODE) solver for the parameter values presented in Table 1. The output, $R(t)$, and its derivatives $\dot{R}(t)$ and $\ddot{R}(t)$, are then fed into Eq. (8) to calculate the scattered pressure. In cases where ode45 failed, the stiff ode15s solver was employed. In the ODE solver, the initial step size was set to 1×10^{-12} , and both the relative and absolute tolerances were set to 1×10^{-6} . The Backward Differentiation Formula (BDF) option was also enabled to improve accuracy when using the stiff solver ode15s. These strict tolerances were chosen to ensure accurate integration of the very stiff system under consideration. The driving acoustic wave is purely harmonic, 30 cycles long, and tapered with a 10 cycles $\tanh$ envelope at both ends to avoid driving transient bubble oscillations. In the Kelvin–Voigt viscoelastic model, the elastic modulus and viscous contributions G and μ , respectively, can be treated independently, see Eq. (1). This paper focuses on the impact of the elastic modulus and the viscosity of the medium was therefore kept the same as for water (1 mPa s), then adjusted by a

Table 1

List of parameters and their values used to run the simulations.

Parameter	Name	Value (min:step:max)	Dimension
R_0	initial bubble radius	(1.0:0.5:5.0)	μm
γ	polytropic exponent (gas)	1.07	[]
$\sigma(R_0)$	initial surface tension (shell)	20	mN/m
κ_s	dilatational viscosity (shell)	6×10^{-9}	kg/s
σ_w	surface tension (water)	72	mN/m
ρ_m	density (medium)	1000	kg/m ³
G	elastic modulus (medium)	(0:10:200)	kPa
μ	viscosity (medium)	2	mPa s
P_0	atmospheric pressure	100	kPa
P_{ac}	driving pressure	(10:10:120)	kPa
f	driving frequency	(0.4:0.2:10.0)	MHz
N_{cy}	total number of cycles	30	[]
N_{env}	number of cycles tapering	10	[]
r	distance from bubble	1	cm

factor of 2 to account for thermal damping [18]. Segers et al. [45] have experimentally measured the surface area-dependent elasticity of the shell $\chi(A)$ from high-precision acoustic attenuation measurements on monodisperse microbubbles. The surface tension curve was then obtained by integrating the elasticity of a range of bubble sizes through $\sigma(A) = \int_A \chi(A) dA$, and subsequently averaging the measured $\sigma(A)$ curves. The result was parametrized using a 10th degree polynomial fit in Matlab, see the Appendix in Segers et al. [45].

3.2. Data analysis

The data was analyzed using the Fast Fourier Transform (FFT) in MATLAB to extract the frequency spectrum of the bubble oscillations for each bubble size, driving frequency, driving pressure, and elastic modulus. The maximum scattered pressure amplitude at the fundamental frequency P_F around the driving frequency was then extracted from the FFT to create the fundamental resonance curves. The same strategy was used to extract the pressure amplitude P_{2H} of the pressure scattered at the second harmonic and P_{subH} of the pressure scattered at the subharmonic frequency.

Fig. 1b shows the pressure $P_s(t)$ scattered by a microbubble with a radius $R_0 = 2.0$ μm , in response to a driving pulse ($f = 2.1$ MHz, $P_{ac} = 120$ kPa) and evaluated 1 cm away from the bubble. The bubble is either embedded in a medium with $G = 0$ kPa (water), or in a medium with $G = 200$ kPa. The frequency spectra of these two examples are displayed in Fig. 1c showing the fundamental response at 2.1 MHz (at the driving frequency) and the higher harmonics near 4.2, 6.3, and 8.4 MHz.

Fig. 1d shows the scatter response as a function of the driving frequency for the fundamental component in the spectrum of a bubble with $R_0 = 2.0$ μm in a medium with $G = 200$ kPa at a range of acoustic driving pressures. When driven at a driving pressure $P_{ac} = 120$ kPa, the resonance frequency in the elastic medium is increased by 1.0 MHz compared to that in water (dashed black line). This is a direct effect of the appearance of the ω^2 dependency in the expression for the scattered pressure, Eq. (9). A stiffer system increases the resonance frequency of the system, and it then also increases the scattered pressure despite the lower oscillation amplitude. While this is not immediately obvious analytically, Eqs. (7) and (13) capture this effect by showing that, owing to this ω^2 factor difference, the radial amplitude decreases with elasticity (Eq. (7)) while the maximum scattered pressure increases with elasticity, with a precisely inverse dependency as compared to the radial oscillation amplitude (Eq. (13)).

Figs. 1e and f show spectrograms of the response of the same bubble to a 120 kPa driving pulse in media with $G = 0$ kPa and $G = 200$ kPa, respectively. Each vertical line in the image is the spectrum of the bubble response to the ultrasound frequency on the horizontal axis. At this pressure, the effect of the shell on the resonance frequency is already

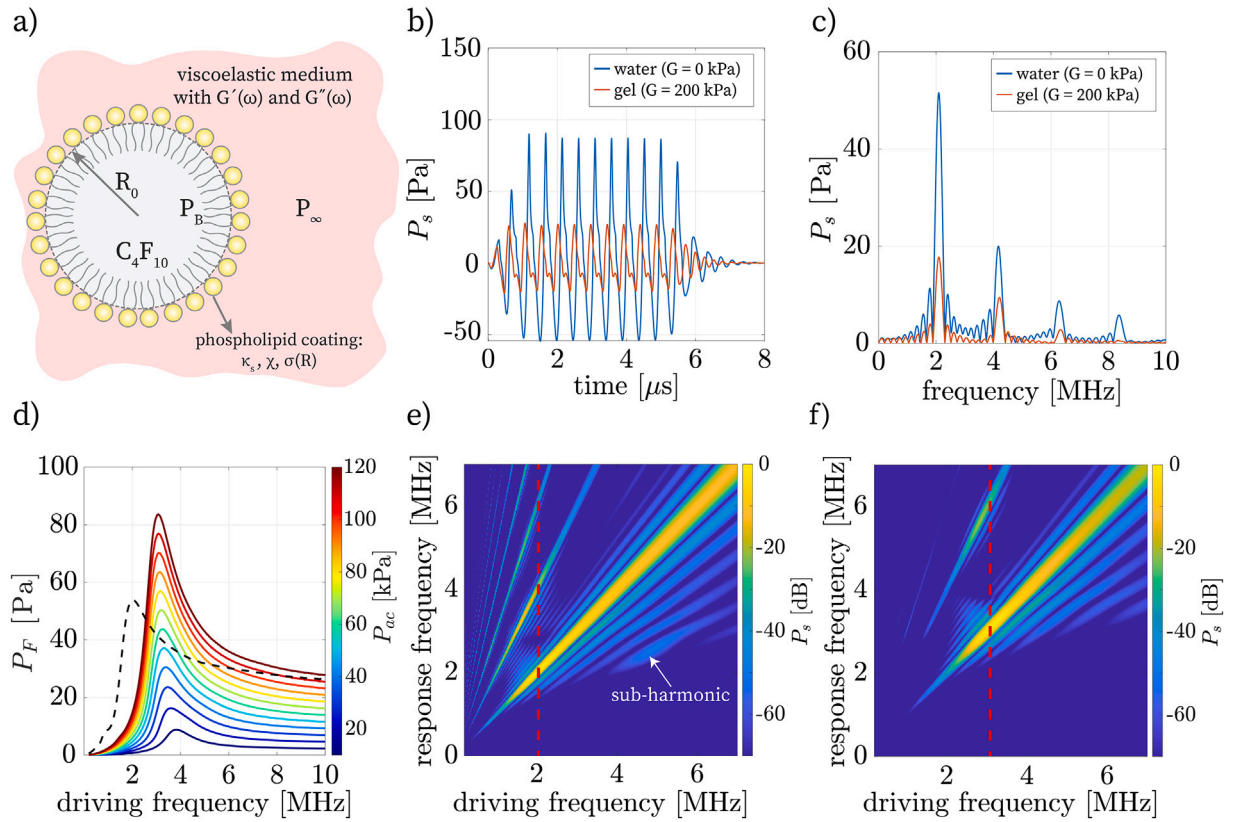


Fig. 1. (a) Schematics of the model: a phospholipid-coated microbubble is embedded in a viscoelastic medium characterized by an elastic modulus G and a viscosity μ . (b) Pressure scattered by a bubble driven at a frequency of 2.0 MHz for $R_0 = 2.0$ μ m, $P_{ac} = 120$ kPa, and $G = 0$ (water) or 200 kPa (gel). (c) Frequency spectra of the curves in b. (d) Fundamental scattered pressure amplitude for $R_0 = 2.0$ μ m and $G = 200$ kPa. The black dashed line is the resonance curve for the same bubble in water driven at a pressure of $P_{ac} = 120$ kPa. (e) and (f) Two-dimensional spectral maps for $R_0 = 2.0$ μ m, $P_{ac} = 120$ kPa, and $G = 0$ kPa (e) or 200 kPa (f). The red dashed lines marks the resonance frequency of the bubble.

minimal, and the changes observed can therefore be attributed to the elasticity of the medium. With these acoustic parameters, the bubble oscillations are small enough to be consistent with the assumption of linear elasticity, see also the Discussion section [37].

4. Results

4.1. Fundamental response

Fig. 2a shows the fundamental resonance curves in terms of radial oscillation amplitude for a $R_0 = 2.0$ μ m microbubble embedded in media with elastic moduli ranging from 0 to 200 kPa. The bubble is driven at a pressure of 10 kPa. The resonance curves are computed from solutions $R(t)$ of the modified Rayleigh–Plesset equation Eq. (1). It is observed that with increasing medium stiffness, the resonance frequency increases, while the oscillation amplitude decreases. This can be understood by plotting the resonance frequency f_r as a function of the resting radius of the bubble, Fig. 2b, also for a driving pressure of 10 kPa. It shows the combined effect of the inverse relationship between the eigenfrequency ω_0 and R_0 , and of the elastic modulus G that increases the eigenfrequency of the bubble, a feature predicted from the linearized Eqs. (3) and (5). While the resonance frequency increases with increasing G , Fig. 2c shows that increasing G leads to a lower oscillation amplitude, a direct consequence of the behavior depicted in the linearized equation Eq. (7).

Fig. 2d shows the fundamental scattered pressure as a function of the driving frequency. These curves are computed by inserting the solutions $R(t)$ and their derivatives into Eq. (8). Increasing medium stiffness leads to both an increased frequency of maximum scatter $f_{m,s}$ and an increased fundamental scattering amplitude P_F . The latter

is fundamentally different from the behavior observed in Figs. 2a–c for the radial oscillation amplitude. Fig. 2e shows the frequency of maximum scatter response as a function of bubble size which, in the limit of low damping, is unaffected by the medium elasticity, see Eqs. (5) and (11). Finally, plotting the maximum scattered pressure as a function of driving pressure gives Fig. 2f, which shows increasing scatter response with increasing medium stiffness, which is in line with Eq. (13). In summary, as expected from Eqs. (7) and (13), a higher medium stiffness leads to a higher resonance frequency, a lower bubble radial amplitude of oscillations, but a larger scattered pressure.

4.1.1. Interplay between shell and viscoelastic medium

As already evidenced in Fig. 1d, for a driving pressure of 120 kPa, the resonance frequency increases by as much as 1.0 MHz when the elastic modulus is increased from 0 to 200 kPa. Furthermore, over that same range of elastic moduli, the maximum scattered pressure increases by up to 55%. Driving a bubble at such pressures entails a non-trivial interplay between the nonlinear viscoelastic shell and the medium stiffness, since the bubble travels out of its elastic regime into the buckling and ruptured regimes. The reduction in effective shell stiffness as the oscillation amplitude increases is further illustrated in Fig. 3.

Fig. 3a shows the frequency of maximum response as a function of driving pressure for a $R_0 = 2.0$ μ m bubble and for different elastic moduli of the medium. The resonance frequency of the bubble is highest at the lowest pressure, where it oscillates purely in the elastic regime of the shell. The medium's elasticity also contributes to the overall stiffness of the system, and the effects of shell stiffness and medium elasticity are of similar magnitude for soft tissues. This is also evident from Fig. 3a, where the resonance frequency of an uncoated bubble

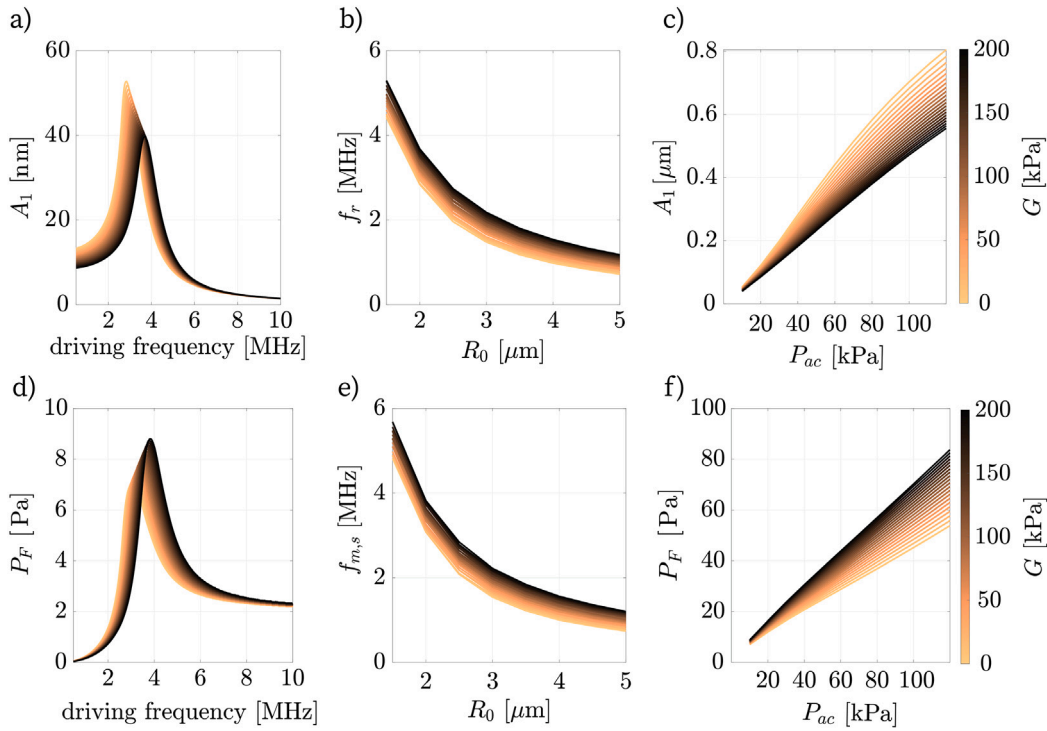


Fig. 2. (a) Fundamental oscillation amplitude A_1 of $R(t)$ at the fundamental frequency as a function of driving frequency for $R_0 = 2.0 \mu\text{m}$ and $P_{ac} = 10 \text{ kPa}$. (b) Resonance frequency of the bubble radial oscillations f_r as a function of R_0 for varying elastic moduli G and $P_{ac} = 10 \text{ kPa}$. (c) Fundamental oscillation amplitude A_1 of $R(t)$ as a function of the driving pressure P_{ac} . (d) Fundamental scattered pressure amplitude P_F as a function of the driving frequency for $R_0 = 2.0 \mu\text{m}$ and $P_{ac} = 10 \text{ kPa}$. (e) Frequency of maximum scatter as a function of R_0 for $P_{ac} = 10 \text{ kPa}$ and for varying elastic moduli G . (f) Maximum scattered pressure as a function of driving pressure P_{ac} for various elastic moduli.

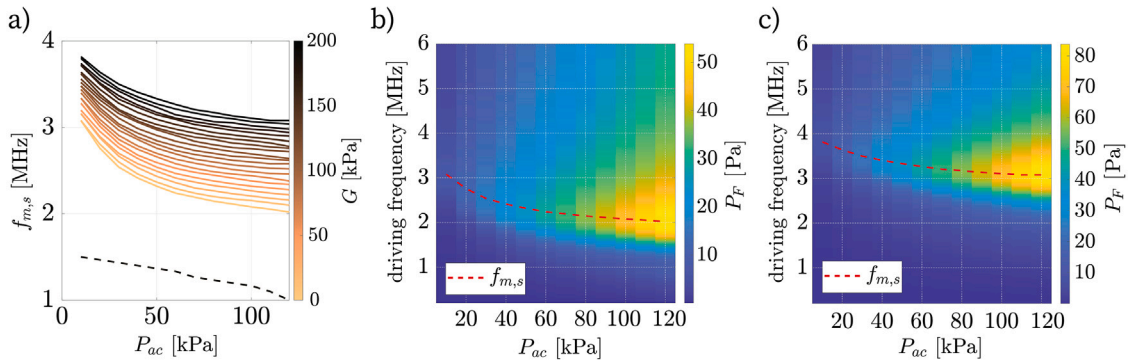


Fig. 3. Interplay between the phospholipid shell and the viscoelastic medium for a bubble with $R_0 = 2.0 \mu\text{m}$. (a) Resonance frequency as a function of driving pressure. The dashed black line corresponds to the resonance frequency of an uncoated bubble in free field. (b) and (c) 2D maps describing the amplitude of the scattered pressure as a function of driving frequency and driving pressure for $G = 0 \text{ kPa}$ and $G = 200 \text{ kPa}$, respectively.

(dotted line) is found to be much lower than that of a coated bubble in water ($G = 0 \text{ kPa}$) which, in turn, is lower by a comparable amount than the resonance frequency of a coated bubble in an elastic material with $G = 200 \text{ kPa}$, owing to the increased stiffness of the system, see Eq. (13). While, at low pressures, shell stiffness has a dominant effect of the resonance frequency of a coated bubble, its importance diminishes at driving pressure above 50 kPa and the role of medium elasticity becomes increasingly important.

The spectrograms in Figs. 3b and c show the corresponding maximum scattered pressure amplitude as a function of driving frequency and pressure for a $R_0 = 2.0 \mu\text{m}$ bubble in media with $G = 0 \text{ kPa}$ and $G = 200 \text{ kPa}$, respectively. The frequency of maximum scattering for each driving pressure is indicated by the dashed line. The distinct difference between both maps highlights the combined effects of driving

frequency, driving pressure, surface elasticity of the shell, and elasticity of the surrounding medium on the scattered pressure.

4.2. Second-harmonic response

As evident from the ordinary differential equation Eq. (1), bubble oscillations are inherently nonlinear. Fig. 1c already shows the emergence of a second harmonic response, as well as that of higher-order harmonics.

Fig. 4a, shows the maximum scattered pressure at the second harmonic for a $R_0 = 2.0 \mu\text{m}$ bubble in a gel with $G = 200 \text{ kPa}$ for different driving pressures. The amplitude of the pressure scattered at the second harmonic is highly dependent on pressure. The dashed line represents the resonance frequency of the same bubble in water at $P_{ac} = 120 \text{ kPa}$. Fig. 4a also shows that the harmonics are not significant

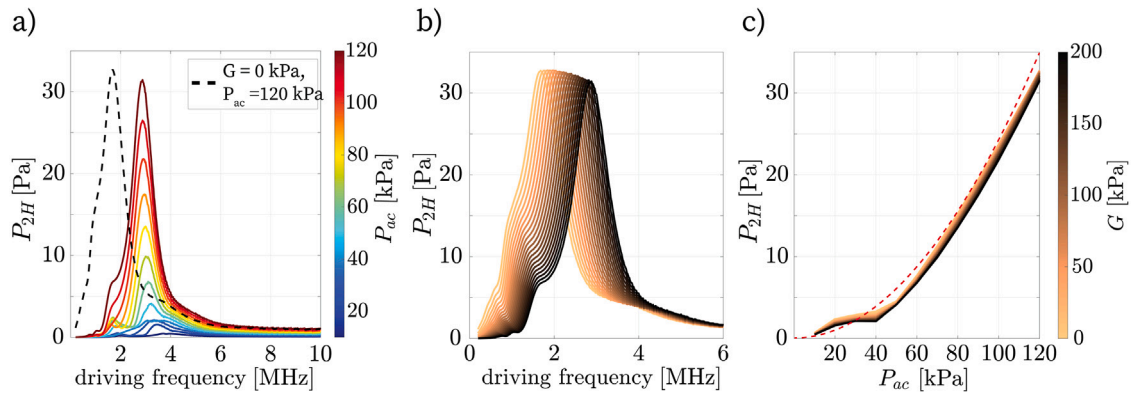


Fig. 4. (a) Maximum scattered pressure at the second harmonic for a bubble with $R_0 = 2.0 \mu\text{m}$ as a function of the driving frequency for different pressures in a medium with $G = 200 \text{ kPa}$. The dashed line represents the resonance frequency of the same bubble in water at $P_{ac} = 120 \text{ kPa}$. (b) Maximum amplitude of the second harmonic for a bubble with $R_0 = 2.0 \mu\text{m}$ in media with different elastic moduli driven at a pressure $P_{ac} = 120 \text{ kPa}$. (c) Maximum amplitude of the second harmonic as a function of the driving pressure for a $R_0 = 2.0 \mu\text{m}$ bubble. The red dashed line is a guide to the eye and shows the quadratic relation of the second harmonic scatter pressure to the acoustic driving pressure.

below a driving pressure of $P_{ac} \approx 40 \text{ kPa}$. Fig. 4b displays the scattered pressure amplitude at the second harmonic for different elastic moduli at a driving pressure of 120 kPa . Similarly to the fundamental response displayed in Fig. 2d, the increased stiffness of the system increases the frequency of maximum response. The amplitude P_{2H} does not vary significantly with increasing elastic modulus, unlike the fundamental response. Specifically, the amplitude decreases by approximately 4% in a medium with $G = 200 \text{ kPa}$ compared to $G = 0 \text{ kPa}$. Fig. 4c shows the amplitude P_{2H} as a function of the driving pressure for increasing elastic moduli. It displays the expected nearly quadratic increase of the harmonic amplitudes above 40 kPa (indicated by the red dotted line to guide the eye) and it confirms the weak dependency of P_{2H} on G .

4.3. Subharmonic response

Subharmonics can appear at two distinct locations in the frequency spectrum: they occur either when the bubble is insonified at its resonance frequency (the so-called TR subharmonic), or when it is driven at twice the resonance frequency, so-called T2R subharmonic [49,50]. TR and T2R subharmonics differ, not only in subharmonic amplitude, but also in stability and non-destructive behavior, in their pressure thresholds, pressure range of active subharmonic emissions and in the strength of ultraharmonics at $\frac{3}{2}f$ [51,52]. For example, the T2R subharmonic can provide a broader non-destructive subharmonic window, whereas the TR subharmonic can yield a stronger but more localized nonlinear response. Fig. 1e has already shown the appearance of a subharmonic response for a $R_0 = 2.0 \mu\text{m}$ bubble driven at an acoustic pressure of 120 kPa in water. There, the T2R subharmonic is visible, but not the TR subharmonic that typically appears at pressures larger than 150 kPa , even for a free gas bubble.

Fig. 5a shows the maximum amplitude of the subharmonic response, P_{subH} , for a $R_0 = 2.0 \mu\text{m}$ bubble suspended in water for acoustic pressures ranging from 10 to 120 kPa . At low pressures, the subharmonic scattering is weak. However, when driven at higher pressures, i.e., $P_{ac} > 50 \text{ kPa}$, the T2R subharmonic rises sharply. Like in Fig. 1e, the T2R subharmonics appear around 4.8 MHz , while the resonance frequency of the bubble is 2.0 MHz , see e.g., Fig. 2b. This mismatch is expected and has been discussed before [50,53]. Throughout the observed pressure range, the TR subharmonic scattered pressure remains low, i.e., below 0.5 Pa .

Fig. 5b shows the same information as Fig. 5a, for the very same bubble, but the bubble is now suspended in a medium with $G = 200 \text{ kPa}$. While, at low driving pressure amplitude, Figs. 5a and b show comparable subharmonic responses, the T2R subharmonic rises more slowly as a function of the driving pressure in the elastic material.

Furthermore, while the TR subharmonic was not visible for the bubble immersed in water, it now reaches an amplitude comparable to that of the T2R subharmonic.

The subharmonic scattered pressure generated by the bubble in the elastic material is approximately half that generated by the bubble in water. To quantify TR and T2R subharmonic behavior, we have plotted the ratio between TR to T2R as a function of G for all driving pressures in Fig. 5c. At pressures below 50 kPa , the TR subharmonic pressure amplitude is comparable to T2R subharmonic pressure amplitude, as shown before. However, at pressures above 50 kPa , the ratio increases monotonically as a function of G .

Fig. 5d shows the maximum subharmonic scattering amplitude as a function of the driving frequency for a $R_0 = 2.0 \mu\text{m}$ microbubble driven at 120 kPa , and for various elastic moduli. Note that the darkest curve in Fig. 5d matches the darkest curve in Fig. 5b. Mirroring the fundamental scattering, the TR subharmonic response increases with increasing medium stiffness until becoming comparable to the T2R subharmonic for $G = 200 \text{ kPa}$. Interestingly, the T2R subharmonic amplitude decreases by about 35% when the elastic modulus increases from 0 to 200 kPa , in sharp contrast with both the fundamental and the TR subharmonic scattering.

To make the effect of the acoustic pressure more explicit, the maximum amplitude of subharmonic scattering as a function of the driving pressure is plotted in Fig. 5e. It shows a clear (and well-known) threshold for the emergence of subharmonics visible for $G = 0 \text{ kPa}$ [23]. In the elastic material we do not observe a clear threshold but rather a transition going from a TR subharmonic-dominated regime towards one dominated by the T2R subharmonic, which shows a grows faster as a function of the acoustic pressure, see Figs. 5b and c.

5. Discussion

For simplicity, we have used a Kelvin–Voigt linear viscoelastic model, which is valid for narrowband excitation, and as long as the medium remains within a linear deformation regime. Gaudron et al. [37] compared the linear Kelvin–Voigt model with non-linear models, e.g., a Neo-Hookean model, showing minimal deviations between these models when bubble expansion remains below $(R_{max} - R_0)/R_0 = \Delta R/R_0 = 0.5$ and bubble compression remains below $\Delta R/R_0 = -0.3$. Since, the maximum relative bubble expansion in our study is $\Delta R/R_0 = 0.4$, we can assume that the Kelvin–Voigt model is a valid approach. Nonlinear scattering in viscoelastic media at larger amplitudes is, however, an interesting avenue to explore.

In our simulations the thermal damping was modeled as an effective viscous correction. However, the way thermal damping is modeled may

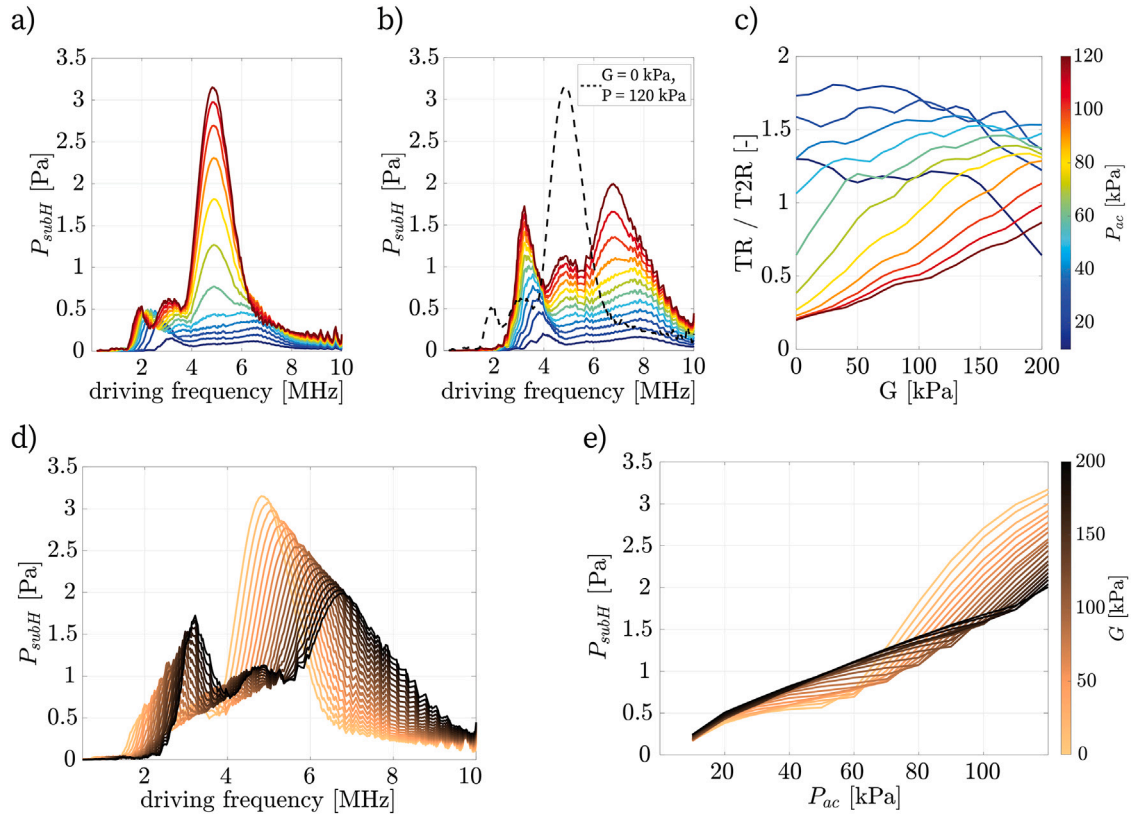


Fig. 5. Maximum amplitude of the subharmonic scatter pressure for a bubble with $R_0 = 2.0 \mu\text{m}$ as a function of the driving frequency in media with (a) $G = 0 \text{ kPa}$ and (b) $G = 200 \text{ kPa}$. (c) Ratio of the TR to T2R subharmonic amplitudes. (d) Maximum amplitude of the subharmonic response as a function of the driving frequency in media with elastic moduli ranging from 0 to 200 kPa, for a bubble with a radius $R_0 = 2.0 \mu\text{m}$ driven at a pressure of 120 kPa. (e) Maximum amplitude of P_{subH} as a function of the driving pressure.

affect the outcome of the simulations, notably the pressure thresholds for the subharmonic response. Prior studies have shown that when thermal damping is modeled more rigorously, the smallest possible pressure threshold for subharmonic generation is not universal, can shift depending on gas properties and excitation frequency and the frequency of maximum subharmonic response may shift. At higher pressures, due to the fast rebound of the bubble, thermal damping becomes weaker compared to viscous damping, and adding a constant thermal viscosity to the liquid viscosity then overestimates the thermal damping [54]. While all these effects may change to some degree the pressure thresholds, the incorporation of thermal damping through an effective viscous correction is a reasonable simplification, especially for perfluorocarbon-filled contrast agents where thermal damping was shown to be relatively weak [54,55] and the qualitative stiffness-dependent trends reported here are expected to remain robust.

The calculations presented in this paper focus on a single microbubble while (medical) applications typically require a suspension of bubbles. Up to a million of monodisperse microbubbles can be contained in the probing volume, even under the assumption of a sparse distribution with a mean bubble-to-bubble distance of at least $500 \mu\text{m}$ to minimize crosstalk and bubble-to-bubble interactions. The response of the collection of bubbles can then easily be detected within the focal volume of a linear array transducer. Such suspensions, however, can never be perfectly monodisperse. A statistical analysis based on typical bubble size distributions would provide insight into the nonlinear scattering of a suspension as compared to that of a single bubble, and thereby shed more light on the practical relevance of the results reported in this study.

These results, especially if combined with the statistical approach described above, pave the road towards the viscoelastic characterization of opaque materials in the MHz range through acoustic scattering. It also shows that the TR/T2R subharmonic ratio increases with elastic modulus for elevated pressures, and this behavior indicates that medium elasticity fundamentally reshapes the nonlinear response landscape of the bubble. To that end it is also interesting to look at the ultraharmonics at $\frac{3}{2}f$ that may fall more conveniently within the bandwidth of the ultrasound transducer. Previous studies have demonstrated that ultraharmonics show strong and diagnostic differences between TR and T2R excitation and often evolve together with the subharmonics [51,52]. The strong dependence of subharmonic scattering on the medium elasticity would potentially allow for using nonlinear imaging, possibly using the TR to T2R ratio in order to measure the elasticity with a high signal-to-background ratio (or contrast-to-tissue ratio for medical application).

Future work may evolve in different ways. First, the interaction of multiple bubbles within the viscoelastic medium can be simulated, as well as the role of bubble size distribution, polydispersity of the bubble suspension and concentration of the bubble suspension that will dictate the bubble-to-bubble distance in the problem [56–58]. Second, the behavior of stabilized microbubbles in a viscoelastic medium can be studied experimentally, for example by a microbubble scattering spectroscopy method that can reveal the behavior of phospholipid-coated microbubbles in e.g. hydrogels at different elastic moduli. These studies can then show how the resonance frequency of monodisperse coated bubbles increases when measured in hydrogels with different storage moduli. Such findings will also provide insights into nonlinear microbubble dynamics within viscoelastic media, contributing to the optimization of contrast-based imaging and therapeutic ultrasound modalities.

6. Conclusion

The scattering behavior of microbubbles is significantly influenced by the stiffness of the surrounding medium. Media with higher stiffness result in higher scattered pressures at the fundamental frequency. For a 2.0 μm bubble, this increase reaches about 50% when the medium elasticity rises from 0 to 200 kPa under a driving pressure of 120 kPa. The shift in resonance frequency induced by the shell and soft tissues is of the same order; however, the effect of the shell diminishes at driving pressures above 70 kPa, while the influence of medium elasticity persists. Under the same conditions, the second harmonic amplitude changes by 5% which, in practice, is negligible. Subharmonic scattering spectra are heavily affected by the presence of a viscoelastic medium. Interestingly, while TR subharmonics become stronger in stiffer media, the T2R subharmonic response decreases with increasing elastic modulus, highlighting the non-trivial influence of the surrounding medium.

CRedit authorship contribution statement

Ali Rezaei: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **David Fernández Rivas:** Writing – review & editing, Supervision. **Guillaume Lajoinie:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Michel Versluis:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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